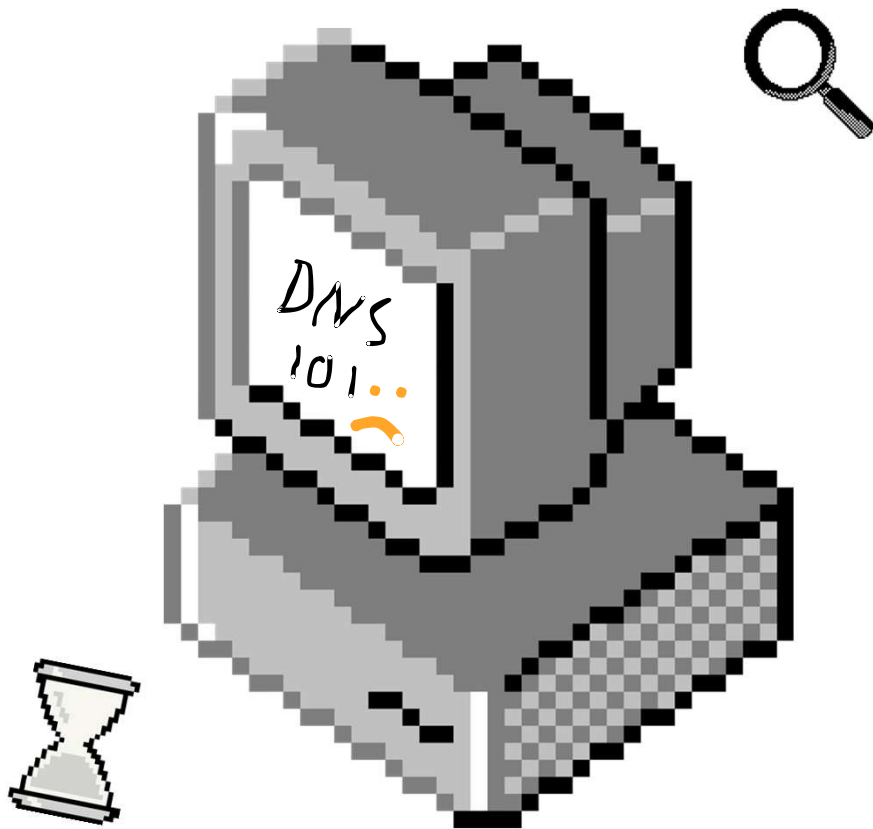


DNS + Cybercrime



@abandonfreewifi / jeffkraken / Jeff Krakenberg

DNS + Cybercrime

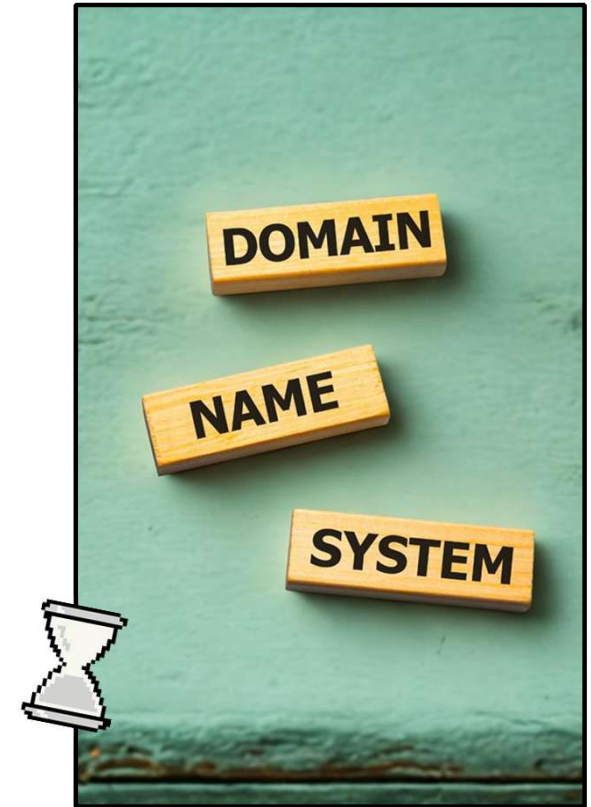


DNS SERVERS & INTERRUPTING CYBER CRIME

@abandonfreewifi / jeffkraken / Jeff Krakenberg

WHAT IS IT?

WHAT DOES IT DO FOR US?



DNS 101

HOW DOES IT WORK?



IF DNS IS SO IMPORTANT, IT'S PROBABLY REALLY SECURE... RIGHT?

Well... No.

Depending on how the system is configured, DNS could allow any to update records. (DNS Poisoning)

But sometimes!

DNSSEC

DNS over TLS or DNS over HTTPS

Restrictive TTLs for DNS Caching

Even Secured DNS isn't perfect

Imagine a perfectly secure system, even if every aspect is locked down, is it still usable?

Tie into Crime

WHY WOULD CRIMINALS TARGET DNS?



Tie into Crime

HOW DO 'CRIMINALS' USE DNS?

TANGENT!

DNS Tunneling is not always active.
It's possible to obfuscate data with base64 encoding, wrap it into small chunks that LOOK like auth keys and then send it off piece by piece to exfiltrate data.



Poisoning

By poisoning the well, cyber criminals can 'infect' a client's cache, forcing them to resolve good names to bad addresses.



Tunnelling

Using queries and responses to exfiltrate data or establish a covert channel for command-and-control communication, often for data theft or to bypass firewalls.



Hijacking

Altering DNS settings (either at the client, router, or registrar level) to redirect users to malicious sites, often used for phishing or malware delivery.



DDoS Amplification (Sounds familiar...)

Exploiting open DNS resolvers to send massive amounts of traffic to a victim, overwhelming target servers and causing denial of service (DoS).

DNS CRIME

ONE IMPRACTICAL EXAMPLE OF MALICIOUS DNS USAGE



Request	Response	Client
 [redacted] Type: TXT, Plain DNS	 Processed 0.17 ms	 ec2-15-184-34-232.me-so BH, Manama Amazon Data
 [redacted] Type: TXT, Plain DNS	 Processed 0.22 ms	 ec2-15-184-34-232.me-so BH, Manama Amazon Data
 [redacted] Type: TXT, Plain DNS	 Processed 0.14 ms	 ec2-15-184-34-232.me-so BH, Manama Amazon Data
 [redacted] Type: TXT, Plain DNS	 Processed 0.21 ms	 ec2-15-184-34-232.me-so BH, Manama Amazon Data

Going back to the NovNov project
that I mentioned at the beginning.



Single Target (Regional Midwest Bank)



Multiple Attack IPs



Requested Information (TXT Record)

DNS CRIME

UNDERSTANDING WHEN THE TARGET IS NOT THE TARGET

```
Served from cache: ✓
Elapsed: 0.17 ms
Response code: NOERROR
Response: TXT: "adobe-sign-verification= [REDACTED] (ttl=2364)
          TXT: "cisco-ci-domain-
          verification=3 [REDACTED] (ttl=2364)
          TXT:
          TXT:
          TXT:
          TXT:
          ip4:1
          ip4:5
          "ip4:
          "ip4:
          ip4:7
          ip4:1
          "ip4:
          ip4:2
          ip4:2
          "ip4:3
          ip4:2
          ip4:2
          "ip4:
          ip4:1
          ip4:1
```

Just imagine this is a HUGE list of IP Addresses associated with a company's email security... because... that's what it was.



DNS CRIME

IF THE TARGET WASN'T THE TXT, WHAT WAS IT?

Process of elimination...



Poisoning

Nothing was updated.
Nothing was forged.



Amplification

Was there a high
volume of traffic?



Tunnelling

Did the data inside
the records change?



Hijacking

Did record ownership
on the system side?

DNS CRIME

DISTRIBUTED SERVICE DISRUPTION

```
.geoip 154.12.61.56  
154.12.61.56  
Hong Kong, Hong Kong
```

Autonomous System	AS979 NetLab Global
Registered Country	United States
Continent	Asia
Country	Hong Kong
City	Hong Kong
Postal Code	999077
Time Zone	Asia/Hong_Kong (UTC+08:00)
Latitude, Longitude	22.2842, 114.1759 (+/- 20 km)

```
.geoip 15.184.34.232  
15.184.34.232  
Bahrain
```

Autonomous System	AS16509 Amazon.com, Inc.
Registered Country	United States
Continent	Asia
Country	Bahrain
Subdivision	Southern Governorate
Time Zone	Asia/Bahrain (UTC+03:00)
Latitude, Longitude	26.1226, 50.5557 (+/- 50 km)

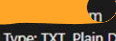
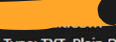
```
.geoip 47.237.97.50  
47.237.97.50  
Singapore
```

Autonomous System	AS45102 Alibaba US Technology Co., Ltd.
Registered Country	United States
Continent	Asia
Country	Singapore
Time Zone	Asia/Singapore (UTC+08:00)
Latitude, Longitude	1.3667, 103.8 (+/- 50 km)

DNS CRIME

HOW TO SPOT MALICIOUS DNS USAGE? (CRIME)



Request	Response	Client
  Type: TXT, Plain DNS	 Processed 0.17 ms	 ec2-15-184-34-232.me-so 📍 BH, Manama 📶 Amazon Data
  Type: TXT, Plain DNS	 Processed 0.22 ms	 ec2-15-184-34-232.me-so 📍 BH, Manama 📶 Amazon Data
  Type: TXT, Plain DNS	 Processed 0.14 ms	 ec2-15-184-34-232.me-so 📍 BH, Manama 📶 Amazon Data
  Type: TXT, Plain DNS	 Processed 0.21 ms	 ec2-15-184-34-232.me-so 📍 BH, Manama 📶 Amazon Data



Unrelated, repeated domain requests

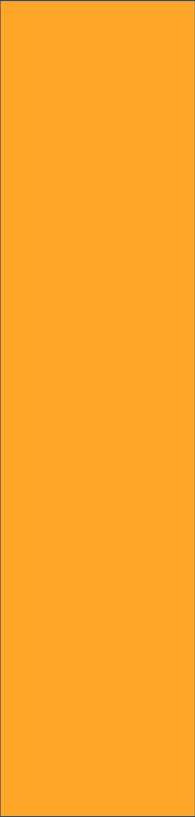
This can hint toward either DNS Tunneling, if the traffic is low, or DNS Amplification of a DDoS attack if the traffic keeps building.



Unknown, unrecognizable hosts

Even one-off requests from unknown IPs or Hosts can be an indicator of something going on. Filtering by GEOIP or Hostname can help identify these requests.

DNS CRIME



18:13:52 11/13/2025	Blocked Custom filtering rules	154.12.61.56	⋮
18:13:52 11/13/2025	Blocked Custom filtering rules	154.12.61.56	⋮
18:13:52 11/13/2025	Blocked Custom filtering rules	154.12.61.56	⋮
18:13:52 11/13/2025	Blocked Custom filtering rules	154.12.61.56	⋮
18:13:52 11/13/2025	Blocked Custom filtering rules	154.12.61.56	⋮
18:14:38 11/13/2025	Blocked Custom filtering rules	ec2-15-184-41-58.me-south-... BH, Manama Amazon Data Services B...	⋮
18:14:37 11/13/2025	Blocked Custom filtering rules	ec2-15-184-41-58.me-south-... BH, Manama Amazon Data Services B...	⋮
18:14:37 11/13/2025	Blocked Custom filtering rules	ec2-15-184-41-58.me-south-... BH, Manama Amazon Data Services B...	⋮
18:14:37 11/13/2025	Blocked Custom filtering rules	ec2-15-184-41-58.me-south-... BH, Manama Amazon Data Services B...	⋮
18:14:37 11/13/2025	Blocked Custom filtering rules	ec2-15-184-41-58.me-south-... BH, Manama Amazon Data Services B...	⋮
18:14:37 11/13/2025	Blocked Custom filtering rules	ec2-15-184-41-58.me-south-... BH, Manama Amazon Data Services B...	⋮
18:14:37 11/13/2025	Blocked Custom filtering rules	ec2-15-184-41-58.me-south-... BH, Manama Amazon Data Services B...	⋮
18:14:37 11/13/2025	Blocked Custom filtering rules	ec2-15-184-41-58.me-south-... BH, Manama Amazon Data Services B...	⋮



NEXT STEPS?



CREATE CUSTOM FILTERING OR 'BLOCK' RULES

- SOURCE OR DESTINATION?
- WHAT'S BETTER?



ALERT AFFECTED PARTIES

- Target
- Hosting Infra.



DOUBLE CHECK YOUR DOCUMENTATION

- CYA (Cover your assets)
- Loop back to make sure you didn't miss anything

TIME CHECK

Look, I may or may not have the ADHD real bad and needed to leave myself a little intermission to check the time.

BYODNSSD - BUILD YOUR OWN DNS SERVER DROPLET

What you'll need:

- Internet Connection
- Digital Ocean Account
- Access to Terminal (Admin in Windows, root in the other stuff)

STEPS:

- 1. Create a Droplet in Digital Ocean (CentOS or Rocky preferred)**
- 2. Configure SSH KEY Access**
- 3. Connect to the Droplet**
- 4. Install AdGuard Home by curl-ing the repo**
- 5. Complete Configs via Web Browser**

Create Droplet

Choose a datacenter region

 Amsterdam • Datacenter 3 • AMS3

▼

VPC network - default-ams3

All resources created in this datacenter will be members of the same [VPC network](#) . They can communicate securely over their Private IP addresses.

Choose a Droplet Plan

☒ Basic
Shared CPU

☐ General Purpose
Dedicated CPU

☐ CPU
Dedicated CPU

CPU Options

☒ Regular
Disk Type: SSD

☐ Premium AMD
Disk Type: NVMe SSD

☐ Premium Intel
Disk Type: NVMe SSD

Select a Plan

☒ \$4.00/mo
1 vCPU - 512 MB RAM - 10 GB SSD - 500 GB Transfer

Choose an image

OS

Marketplace (271)

Custom Images

☐  Ubuntu
RECOMMENDED

24.04 (LTS) x64 ▼

☐  Debian

13 x64 ▼

☐  Fedora
Requires an SSH Key

43 x64 ▼

☐  CentOS

9 Stream x64 ▼

☐  AlmaLinux

AlmaLinux 9 ▼

☒  Rocky Linux
Requires an SSH Key

9 x64 ▼

Authentication

SSH Keys

Password

Add an SSH Key for authentication

An SSH key pair is a more secure way to connect to your Droplet

Add SSH Key

Copy your public SSH key and paste it in the space below. For instructions on how, follow the steps on the right.

SSH Key content*

ssh-ed25519

AAAAB3NzaC1naW4...

Give your SSH Key a name*

DODEMO4BSKNOX

Cancel

Add SSH Key

Follow these instructions to create or add SSH keys on Linux, MacOS & Windows. Windows users without OpenSSH [can install and use PuTTY instead](#)

Add a new key

Copy an existing key →

Step 1: Generate an SSH Key

1. Open your **PowerShell** (use regular mode, not administrator)

2. Run this command

```
ssh-keygen
```

3. Run this command

- Enter file in which to save the key: Press Enter to use the default filename and

Give your Droplet a name

Can only contain alphanumeric characters, dashes, and periods only.

adbsk

Finalize

Select a project



NovNov



Summary



Compute

\$4.00/mo

Plan Type: Basic
CPU Option: Regular
vCPU: 1
RAM: 512 MB
Disk: 10 GB
Bandwidth: 500 GB
Slug: s-1vcpu-512mb-10gb

Total cost

\$4.00/month

\$0.006/hour

Create Droplet

CONNECTION DETAILS

Public IPv4

161.35.159.53

Copy



```
PS C:\Users\JeffKrakenberg> ssh root@161.35.159.53
The authenticity of host '161.35.159.53 (161.35.159.53)' can't be established.
ED25519 key fingerprint is SHA256:3srqHANfUjgmzurz90f9ZwCYN2H3Gb8mSASs1
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '161.35.159.53' (ED25519) to the list of known hosts.
root@adbsk:~#
```

github.com/AdguardTeam/AdGuardHome

github.com/AdguardTeam/AdGuardHome

[README](#) [Contributing](#) [GPL-3.0 license](#) [Security](#)

Getting Started

Automated install (Linux/Unix/MacOS/FreeBSD/OpenBSD)

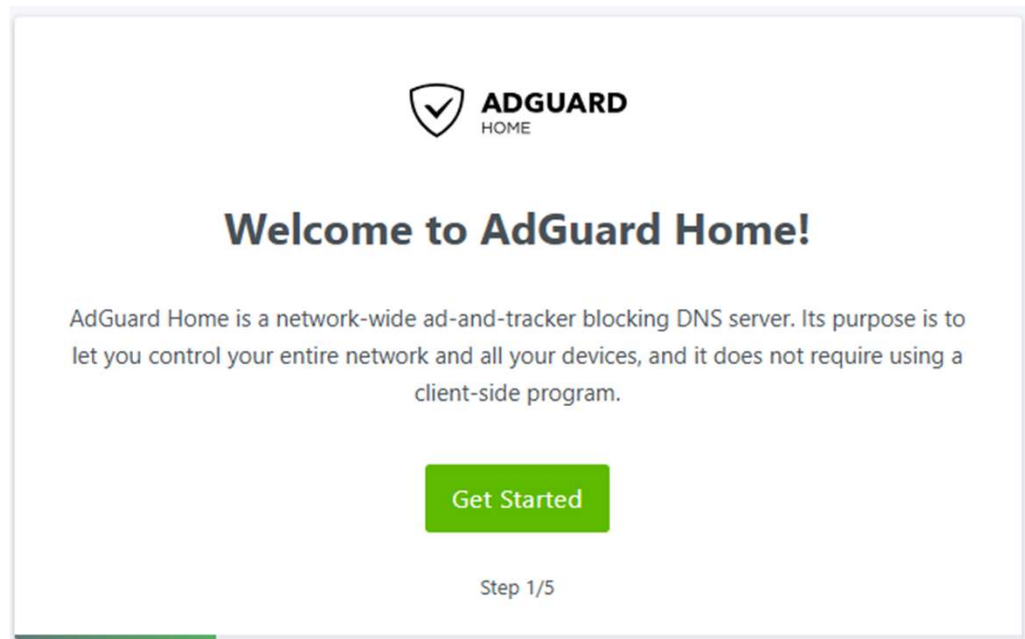
To install with `curl` run the following command:

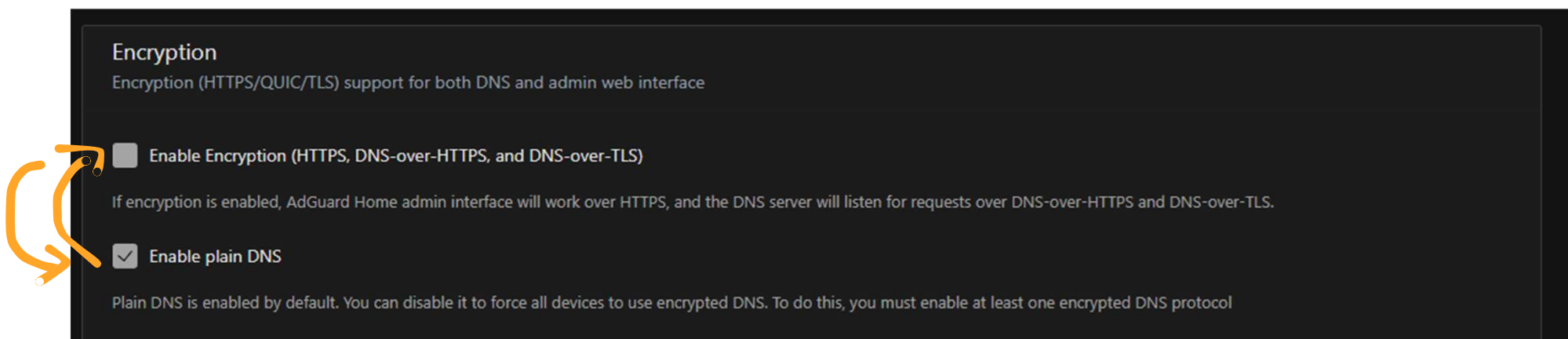
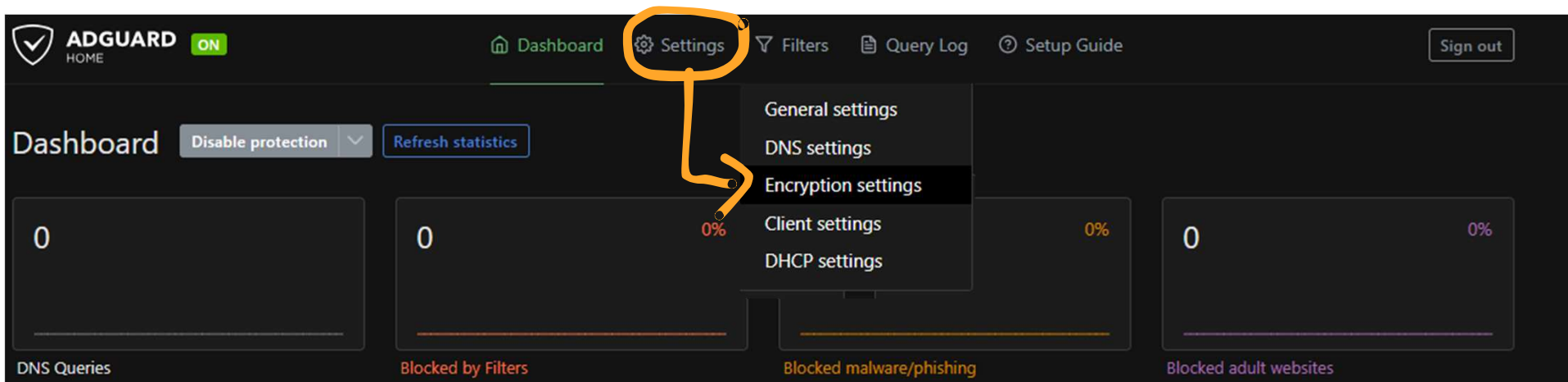
```
curl -s -S -L https://raw.githubusercontent.com/AdguardTeam/AdGuardHome/master/scripts/install.sh | sh -
```

```
AdGuard Home is now installed and running  
you can control the service status with the following commands:  
sudo /opt/AdGuardHome/AdGuardHome -s start|stop|restart|status|install|uninstall  
[root@adbsk ~]# |
```

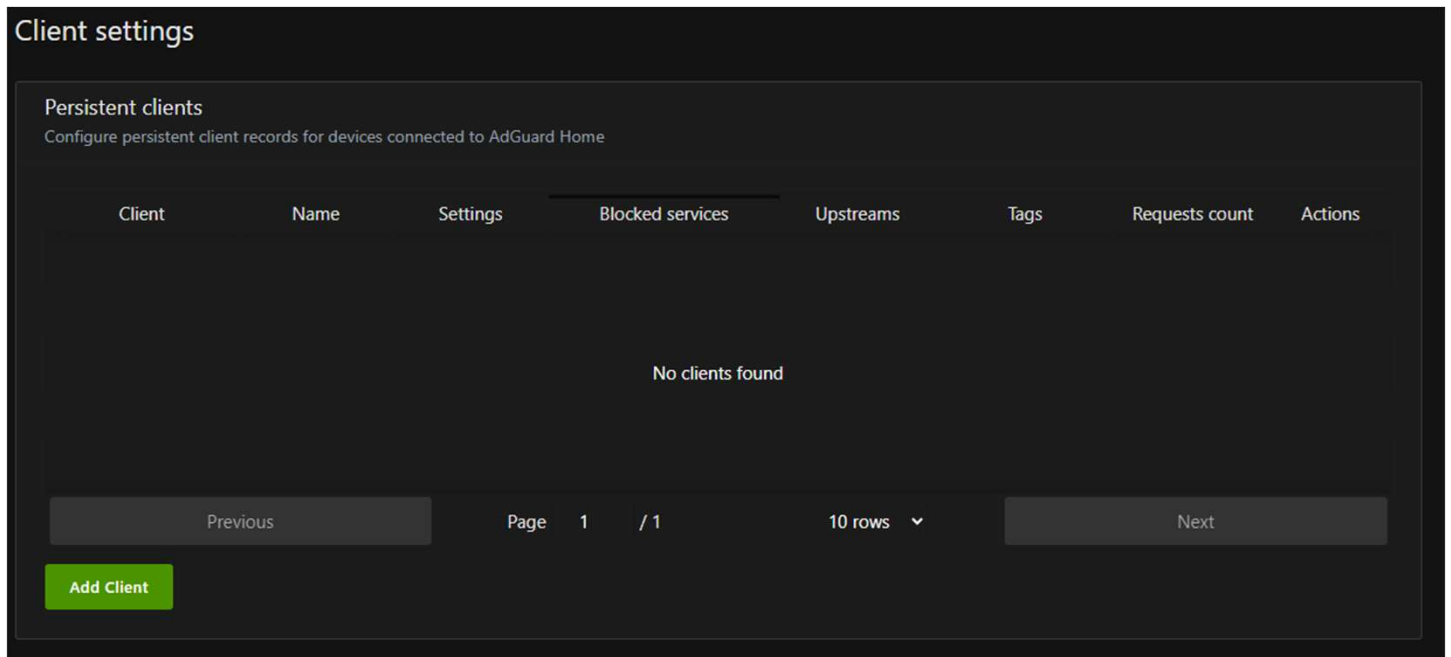
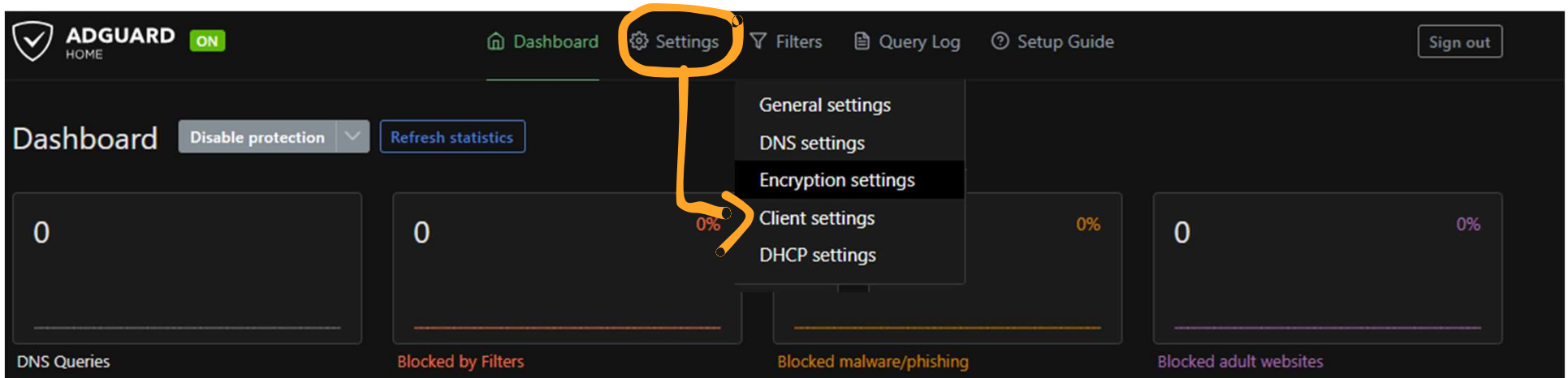
In your preferred browser:

- <https://{droplet's-IP-Address}:3000>





flip the defaults to force
encryption



**Configure persistent clients
for your known devices**

Block all others.

BYODNSSD - BUILD YOUR OWN DNS SERVER DROPLET

Understand that this is JUST THE START.

As your droplet DNS server is now, it can still be abused by malicious users. To prevent being the DITM (Dummy in the middle), consider the following:

- Disable Open Recursive Resolution
- Enable DNSSEC Validation
- Enable Response Rate Limiting (RRL)

THANKS!

THIS WAS A LABOR OF LOVE AND
HUMOR TO HELP WRAP OUR BRAINS
AROUND DNS MISUSE AND ABUSE.

ADDT'L QUESTIONS OR CONNECTIONS
LINKEDIN - IN/ABANDONFREEWIFI
EMAIL - JEFFK@CYBERGOBLIN.ORG